



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SURYA FUEL AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

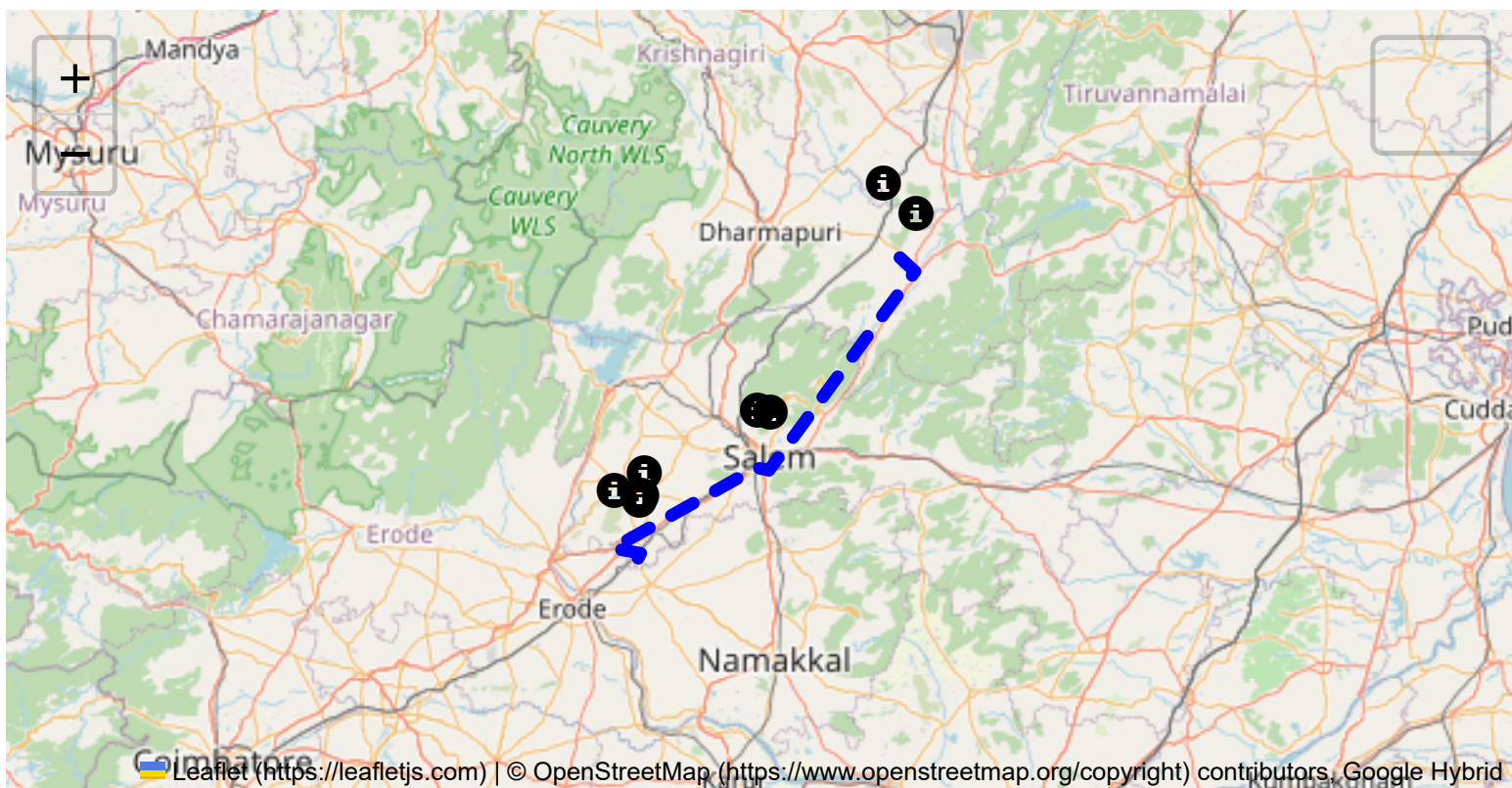
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

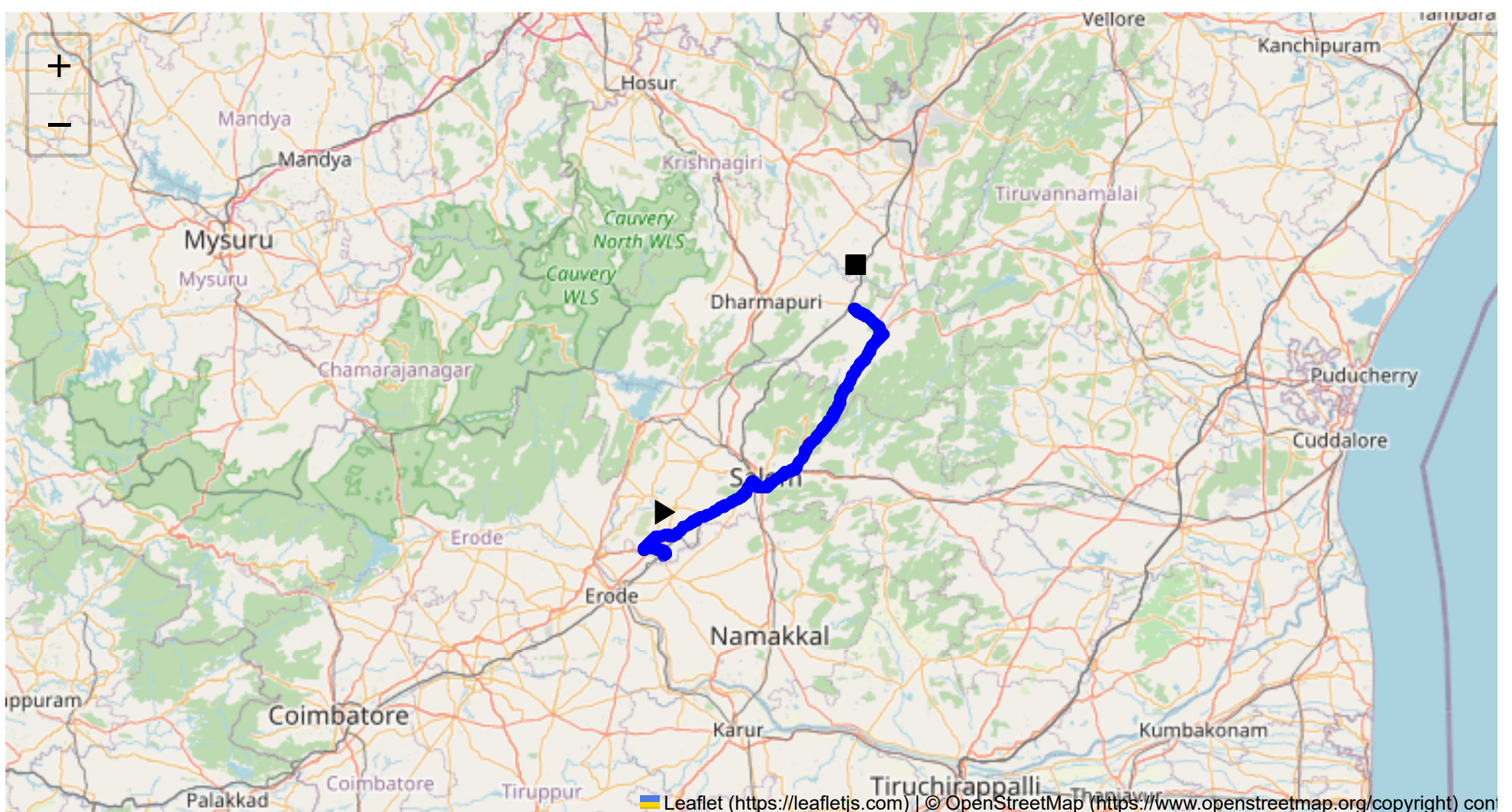
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 128.74 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.7 hours
Start: (11.4381, 77.8734)
End: (12.11655181, 78.40353772)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route stretches approximately 128.74 kilometers from Sangagiri to Morappur via CVXF+HPR, Bhavani Main Road, Salem - Kochi Highway, Kondalampatti Bypass, Attur Main Road, and Harur. This route travels primarily through rural and semi-urban areas of Tamil Nadu, encompassing highways well-trafficked by local and regional commuters.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** The region typically experiences tropical weather. During summer months (March to June), high temperatures can reach 35-40°C. The monsoon season (June to September) brings heavy rains, while the winter months (November to February) are cooler with temperatures ranging from 20-25°C.
- **Potential Hazards:** Heavy rainfall during the monsoon can lead to flooding and reduced road visibility, increasing the chance of hydroplaning. High temperatures in summer may cause road and vehicle stress.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (8 AM - 10 AM) and evening (5 PM - 8 PM) on weekdays typically see peak traffic, especially around urban centers like Salem.
- **Congestion-Prone Areas:** Salem's Kondalampatti Bypass and Bhavani Main Road are especially prone to congestion, particularly near market areas and industrial zones.

4. Assessment of Road Quality and Infrastructure

- **Highways:** Generally well-maintained, especially major sections like the Salem-Kochi Highway.
- **Local Roads:** Some rural roads can be narrow and lack proper signage and lighting, posing risks for large vehicles.

5. Suggestions for Alternative Routes for Emergencies

- In case of roadblocks or accidents, consider diverting through NH44 for a section before merging back on the primary route near Kondalampatti. Additional detours via smaller state highways can bypass Salem city congestion.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits:** Ensure appropriate permits for transporting hazardous materials.
- **Timings:** Heavy vehicles may face restrictions in certain city areas during peak hours.
- **Safety Compliance:** Regularly check spill management equipment and adhere to safety protocols mandated by the Tamil Nadu State Transport Authority.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Incidents:** Data indicates sporadic incidents involving overturned trucks and spillage primarily due to poor weather and road conditions.

- **Nature of Incidents:** Most incidents resulted from overspeeding or improper handling of vehicle loads.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Zones:** Crosses through some agricultural regions; potential risk of chemical contamination in case of spills.
- **Natural Habitats:** Minimal; however, awareness around water bodies during monsoons is necessary to avoid spillage into local lakes or streams.

9. Analysis of Communication Coverage

- **Coverage:** Generally good mobile network availability in urban and semi-urban areas; some rural stretches might experience weak signals.
- **Potential Dead Zones:** Rural areas beyond Harur could have intermittent coverage.

10. Estimated Emergency Response Times

- **Urban Areas (Salem):** Approximately 20-30 minutes for emergency services due to proximity of hospitals and fire stations.
- **Rural Areas:** Response time may extend to 60 minutes or more owing to distance and terrain.

11. Overall Summary of Risk Assessment

The route presents moderate risks primarily from traffic congestion, weather impacts, and some rural road conditions. Attention to weather forecasts, especially in monsoon, and adherence to safety standards can mitigate risks. Well-prepared emergency response plans should be in place, considering potential delays in certain segments. Regulatory compliance and alternative routes can ensure safer transportation of hazardous materials. Overall, while certain sections like those through Salem are better equipped, rural segments require careful navigation and monitoring.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45543, 77.81740	30 KM/Hr
9	Turn	High	11.45631, 77.81668	15 KM/Hr
10	Turn	Medium	11.53569, 77.95787	30 KM/Hr
11	Turn	Medium	11.64120, 78.11797	30 KM/Hr
12	Turn	High	11.64216, 78.11875	15 KM/Hr
13	Turn	High	11.64162, 78.12160	15 KM/Hr
14	Turn	Medium	11.64087, 78.12157	30 KM/Hr
15	Turn	High	11.63948, 78.12081	15 KM/Hr
16	Turn	Medium	11.62821, 78.13348	30 KM/Hr
17	Turn	Medium	11.62836, 78.16483	30 KM/Hr
18	Turn	High	11.66380, 78.20648	15 KM/Hr
19	Turn	High	11.67120, 78.23136	15 KM/Hr
20	Turn	High	11.67190, 78.23144	15 KM/Hr
21	Turn	Medium	11.67429, 78.23896	30 KM/Hr
22	Turn	Medium	11.80416, 78.33319	30 KM/Hr
23	Turn	Medium	11.81219, 78.34077	30 KM/Hr
24	Turn	Medium	11.83472, 78.35285	30 KM/Hr
25	Turn	Medium	11.99572, 78.44559	30 KM/Hr
26	Turn	Medium	12.01548, 78.45650	30 KM/Hr
27	Turn	Medium	12.04309, 78.48361	30 KM/Hr
28	Turn	Medium	12.04429, 78.48380	30 KM/Hr
29	Turn	Medium	12.04490, 78.48355	30 KM/Hr
30	Turn	High	12.04984, 78.47480	15 KM/Hr
31	Turn	High	12.06014, 78.47787	15 KM/Hr
32	Turn	High	12.11584, 78.40337	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
14	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
15	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
16	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
17	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
18	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
19	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
20	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
21	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
22	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
23	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
27	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
28	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
36	hospital	Government Hospital, Harur	12.061178, 78.4756913	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
37	hospital	Subam Hospital	12.0595128, 78.4766713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
13	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
24	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
25	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
29	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
30	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
31	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
32	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
33	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
34	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
35	school	Government Girls Hr Sec School	12.0488709, 78.4762447	30 km/h	Medium
38	college	Government Arts and science college,Harur	12.0621981, 78.4738033	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53569, 77.95787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64120, 78.11797



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64216, 78.11875



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64162, 78.12160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64087, 78.12157



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62821, 78.13348



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62836, 78.16483



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66380, 78.20648



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67120, 78.23136



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67190, 78.23144



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67429, 78.23896



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.80416, 78.33319



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81219, 78.34077



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83472, 78.35285



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99572, 78.44559



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01548, 78.45650



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04309, 78.48361



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04429, 78.48380



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04490, 78.48355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.04984, 78.47480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.06014, 78.47787



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.11584, 78.40337

